use golden search to find minima

$$f(x) = \frac{x^2}{a} - b\sin x$$

$a \to$ 10 to 15 int.
$b \to$ 2 to 3 int.

Take golden ratio $\phi = 1.618$

Take interval $x_L = 0$ to $x_U = 4$

Iteration 1

$$d = (\phi - 1)(x_U - x_L) = \boxed{}$$

$$x_1 = x_L + d = \boxed{}$$

$$x_2 = x_U - d = \boxed{}$$

$$f(x_1) = \boxed{} \qquad f(x_2) = \boxed{}$$

If $f(x_1) < f(x_2)$

New values

$x_L = \boxed{}$ $\qquad x_L = x_2$

$x_U = \boxed{}$ $\qquad x_U = x_U$

else

New values

$x_L = \boxed{}$ $\qquad x_L = x_L$

$x_U = \boxed{}$ $\qquad x_U = x_1$

Iteration 2

$$d = (\phi - 1)(x_U - x_L)$$

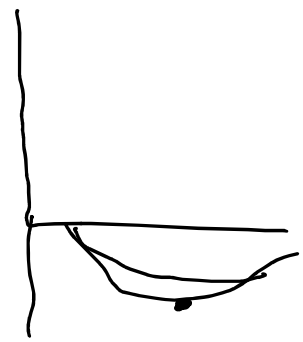
$$x_1 = x_L + d$$

$$x_2 = x_U - d$$

$$f(x_1) = \boxed{} \qquad f(x_2) = \boxed{}$$

If $f(x_1) < f(x_2)$

New values

If $f(x_1) < f(x_1)$

iter	x_L	x_U	$x_1 = x_L + d$	$x_U - d$ x_2	$f(x_1)$	$f(x_2)$	d
1	0	4					
2	1	4					
⋮							
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